Drake's Collaboration Networks

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AI Disclosure

I used Claude, an AI assistant from Anthropic, to help create the data sets. Claude cleaned the Spotify track list, standardized artist names, assigned tracks to projects, calculated tie weights and node positions, and built the node and edge lists. I provided the track list and reviewed the results against Spotifys track list.

Introduction

This project uses two networks to show how Drake's collaborations have changed over time. The first looks at guests on his albums. The second looks at his guest appearances on other artists' songs.

Step 1: The Networks

Album network. The nodes are 21 Drake projects, including albums, mixtapes, EPs, and a compilation, and the 75 artists credited on them. A tie means an artist is credited on that project. Tie strength is the number of credited tracks. Drake is not a node because he would connect to every artist.

Guest network. The nodes are years and the 104 artists whose songs include Drake as a guest. A tie means Drake appeared on that artist's song in that year. Tie strength is the number of songs that year.

Both are affiliation networks, because ties connect artists to albums or years rather than directly to each other.

Step 2: Encoding the Networks

I encoded each network as a node list and an edge list in CSV files. The album network uses `drake_nodes.csv` and `drake_edges.csv`. The guest network uses `drake_guest_nodes.csv` and `drake_guest_edges.csv`.

Both networks come from a 568-track fan-made Spotify playlist of Drake's songs and features. For the album network, I matched Drake-led tracks with guest credits to 21 projects released from 2009 to 2026. I counted every credited artist as a guest and treated Future on *What a Time to Be Alive*, 21 Savage on *Her Loss*, and PARTYNEXTDOOR on *$ome $exy $ongs 4 U* as co-leads. I excluded standalone singles and soundtrack songs. This left 75 guest artists and 116 ties.

For the guest network, I treated the first listed artist as the host when that artist was not Drake. I excluded *$ome $exy $ongs 4 U*, repeated remixes by the same host, and one track credited only to "Various Artists." This left 194 guest songs.

Before reading the data into R, I exported the playlist's track list with Spotlistr, added release years, and separated each track into lead artist, featured artists, title, and year. I standardized artist names and added Drake to four tracks that named him in the title but not in the credits. I then calculated tie weights, song counts, and each artist's average year or project position. I also added label rows for the scale.

Step 3: Reading the Networks into R and iGraph

I read each node list and edge list into R with `read.csv()`. I then used `graph_from_data_frame()` to convert each pair of files into an iGraph network. The edge list goes in the `d` argument, and the node list goes in the `vertices` argument. Using a node list keeps artists' details, such as their type and song count, attached to each node, so I could use them later to set colors, sizes, and labels.

```
library("igraph")

#Load and convert the album network
album_nodes <- read.csv("drake_nodes.csv", header=T, as.is=T)
album_links <- read.csv("drake_edges.csv", header=T, as.is=T)
album_net <- graph_from_data_frame(d=album_links,
                                   vertices=album_nodes,
                                   directed=F)

#Load and convert the guest network
guest_nodes <- read.csv("drake_guest_nodes.csv", header=T, as.is=T)
guest_links <- read.csv("drake_guest_edges.csv", header=T, as.is=T)
guest_net <- graph_from_data_frame(d=guest_links,
                                   vertices=guest_nodes,
                                   directed=F)
```

Both networks are undirected because a tie simply links an artist to a project or year. The album network has 21 projects, 75 artists, and 116 ties. The guest network has 21 years, 104 artists, and 165 ties, with weights covering all 194 guest songs. Both node lists also include label rows for the scale, which are not counted here. I checked each network by printing `V()` and `E()` before plotting.

Steps 4: Plotting and Visualizing the Networks

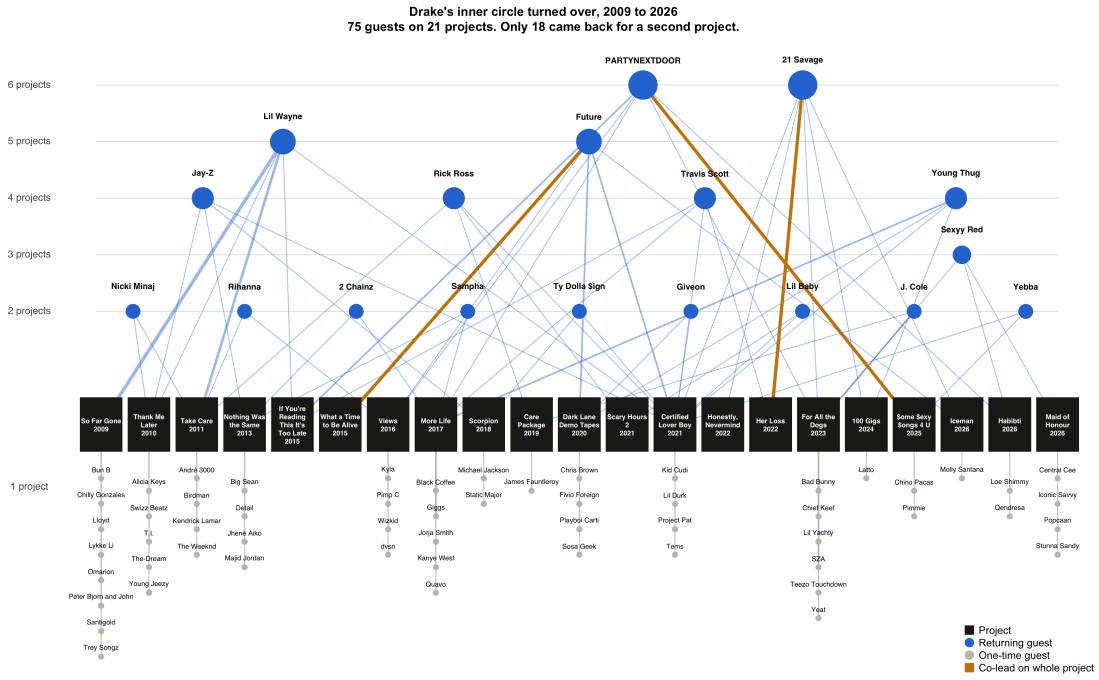


Figure 1: Guests on Drake's projects, 2009 to 2026.

I arranged both graphs as timelines so they show change over time. Albums or years run from left to right. Artists with more appearances sit higher, and one-time artists are stacked below their album or year. Larger dots show more appearances, wider lines show stronger ties, and grid lines make the levels easier to read.

In the album network, blue marks returning guests, gray marks one-time guests, and gold marks a co-lead on a full album.

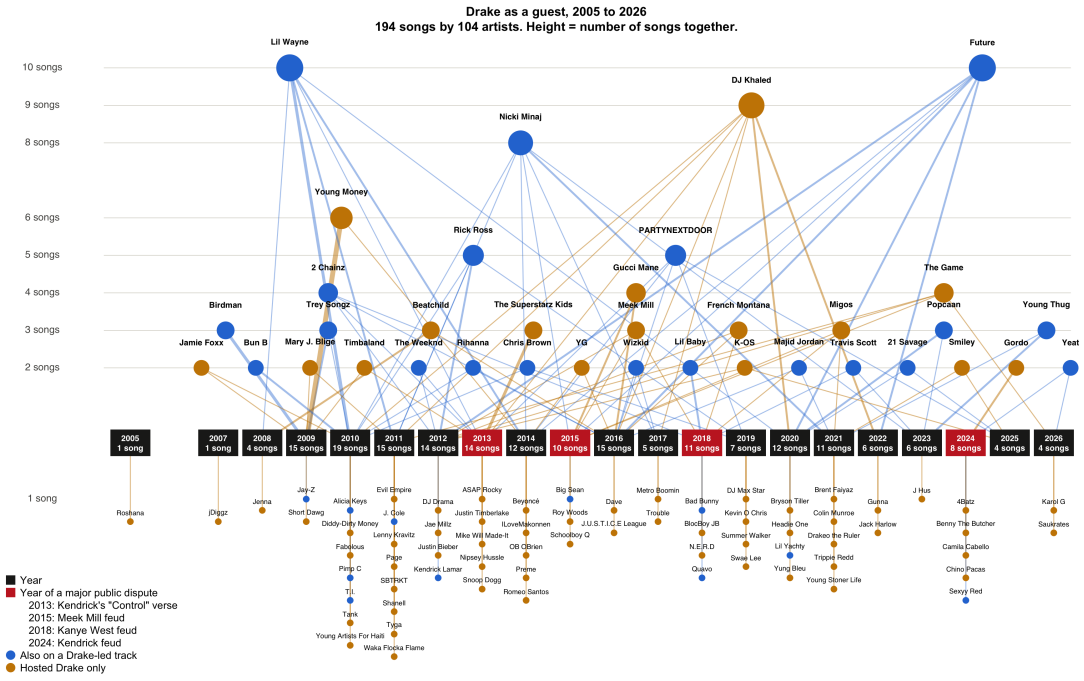


Figure 2: Drake's guest appearances, 2005 to 2026.

In the guest network, blue marks hosts who are also credited on a Drake-led track, and gold marks artists who only hosted Drake. Red year boxes mark four major public disputes: Kendrick Lamar's "Control" verse in 2013, the Meek Mill feud in 2015, the Kanye West feud in 2018, and the Kendrick Lamar feud in 2024. Rows with several artists are spaced evenly in time order, so the lines show the actual years.

Summary

Figure 1 shows that Drake's project guests change over time. Only 18 of the 75 guests returned for a second project. Jay-Z and Lil Wayne connect many of the earlier projects, while 21 Savage and Sexyy Red appear only in later ones. PARTYNEXTDOOR bridges both periods, and he and 21 Savage appear on the most projects, with 6 each.

Figure 2 shows that Drake's guest work was concentrated early in his career. The years 2009 to 2016 account for 114 of the 194 guest songs, with a peak of 19 in 2010. Lil Wayne and Future each have 10 songs, followed by DJ Khaled with 9 and Nicki Minaj with 8. Only 36 of the 104 hosts worked with him more than once, so most of these ties were one-time collaborations.

Collaborations and Public Feuds

Drake fans know the beefs, but the playlist shows the timing behind them. The bridges often fell long before the diss tracks did. The Weeknd's "Live For" and ASAP Rocky's "F**kin' Problems" date to 2013, more than a decade before both sided against Drake in 2024. Rick Ross shared 15 tracks with Drake, but "You Only Live Twice" in 2021 was the last, three years before "Push Ups." Kendrick Lamar's ties ended in 2013, the year of his "Control" verse, and never came back. Still, beef isn't always final. Meek Mill went from "Back to Back" to "Going Bad," Chris Brown returned with "No Guidance," and Future came back on "Ran to Atlanta" in 2026 after a gap in shared tracks from 2023 to 2025.

These patterns suggest that a quiet period in the network can come before a public feud, but they do not show that feuds ended these collaborations. I selected these artists because their conflicts were publicly reported, and private relationships may have changed earlier.

Limitations

The years in the guest network are Spotify release dates, so some older songs appear later than their original release. The playlist is fan-made, and Drake is not listed in the credits of 23 tracks. The album network covers 21 projects and leaves out 10 standalone singles and soundtrack songs. I assigned tracks to projects by hand because the playlist does not include album names.

Sources

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